Alina Du

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Education

Columbia University May 2028
B.S. Computer Science; Minors in Financial Engineering, Math | GPA: 3.9/4.0 New York, NY

- **Coursework:** Machine Learning, Optimization, Software Quality and Design, Data Structures
- **Activities:** Teaching Assistant; Columbia Lioness Quantitative, WiCS, SWE

Awards & Skills

Awards: Jane Street WiSE (top 60); MIT Math Prize for Girls; 2x AIME Qualifier; 8x Waterloo Math Contest Top 1%; Waterloo National Math Scholarship (15 recipients)
Languages: Python, Java, TypeScript, JavaScript, SQL, HTML5
ML/Data: PyTorch, Hugging Face, PEFT/QLoRA, Scikit-learn, NumPy, Pandas, SciPy
Frameworks: React/React Native, Next.js, FastAPI, LangGraph
Backend/Tools: AWS, Google Cloud, PostgreSQL, MongoDB, Git, Linux, GitHub Copilot

Experience

Software Engineer Intern Oct 2026 - Present
Model OS New York, NY

- Building an AI OS platform that fine-tunes private models to keep enterprise IP self-hosted
- Fine-tuning QLoRA lab-report extractors on synthetic NHANES data for on-prem healthcare use

Data and ML Researcher Oct 2026 - Present
Columbia Plasma Physics Lab New York, NY

- Built a Codabench prototype for a CASP-style plasma prediction challenge on ∼15K DIII-D tokamak shots, with a custom Python scoring program and hidden answer keys
- Developing latent-space analysis of sparse data regions to predict out-of-distribution shots

Research Assistant - Data Engineering Sep 2026 - Present
Columbia Experimental Laboratory for the Social Sciences New York, NY

- Built the study UI and GPT-5 mini chat for econ-LLM experiments with 150 conditions
- Built a Python pipeline matching 155 OpenRouter models to Epoch AI capability scores

AI Research Intern May 2026 - Sep 2026
Columbia University New York, NY

- Raised reliability from 91.8% to 96.5% on 50 reasoning trees with risk-targeted verification
- Benchmarked 8 LLMs on 14K+ claims; most stayed 66–95% confident on wrong answers
- Presented research poster at the Jane Street and Columbia Research Fair

Software Engineer Intern Mar 2026 - Jun 2026
Intelligence Cubed New York, NY

- Built a single backend API that lets users deploy AI models to AWS, GCP, and Cloudflare
- Wrote a Node.js proxy routing Anthropic API calls to Cloudflare-hosted Kimi K2 for model swaps

Software Engineer Intern Aug 2025 - Mar 2026
MeloMed - Health Tech Startup New York, NY

- Reduced infrastructure costs by 23% through serverless AWS deployment and caching
- Cut PostgreSQL query latency from ∼800 to ∼150 ms via indexing and query rewrites

Projects

Lionplan.ai | Next.js, TypeScript, Python, Postgres, Claude API

- Launched an AI course planner used by 1,000+ students, with 4-year planner and conflict detection
- Built a tool-calling AI agent that adds, drops, and swaps courses, validated against prereq rules
- Built Python scrapers and an LLM pipeline parsing 30K+ reviews, 1.3K+ prereqs and 4.6K+ courses

Pac-Manhattan | JavaScript, Three.js, Python, Shapely, Gemini API, ElevenLabs

- Rendered 171K NYC buildings in 3D with Three.js and A* pathfinding over a 27K-edge street graph
- Wrote a Python/Shapely pipeline turning OpenStreetMap and aerial imagery into per-borough maps
- Added a Gemini pipeline that checks riddles for leaked answers, with ElevenLabs narration